1 Socio-Pragmatic Architecture and System Component Specifications

1.1 Pragmatics, Level-k Reasoning, and Searlean Realities

To model social networks where agents think about other agents' thoughts, a custom Probabilistic Domain Specific Language (DSL) was developed. This language parses recursively nested mental states (`thinks`, `chooses`, `knows`, `observes`) to simulate Rational Speech Act (RSA) matrices and level-k strategic convergence (e.g., Schelling games, political and religious belief polarization networks).

To satisfy John Searle's philosophy of the social world, the language was expanded to Grammar 3, embedding a dedicated, constitutive status operator:

$$\mathcal{D}_{\text{inception}}\Big(\langle C,\, X,\, Y\rangle\Big) \implies (X \text{ counts as } Y \text{ in } C) \tag{1}$$

This allows the engine to instantiate institutional facts that cannot be measured via physical sensory inputs. The grammar cleanly separates illocutionary forces into five distinct categories: Assertives, Directives, Commissives, Expressives, and Declarations. This framework manages security perimeters, access control, and legal obligations containing strict temporal deadlines (t_{deadline}).

1.2 Specification and System Components

The neuro-symbolic system acts as a bounded, type-safe transducer. It inputs structural strings or queries and produces learned contexts, classifications, or deductive responses based on the available training material.

The system is powered by seven interlocking components:

1. **LLM as Co-Designer:** Acts as an automated reward stencil synthesizer and meta-level instructor.
2. **Python Orchestrator:** The master continuous execution plane and virtual machine runner.
3. **Forest of Decision Trees:** The core associative, hierarchical memory array for learning situational contexts.
4. **The Prolog Guardrail (`kb.pl`):** The formal logical attention filter ensuring type-safe branch splits.
5. **RL with DSLs:** Sequential decision-making engines optimizing policies over bounded continuous languages.
6. **Automated Algebraic Solver:** Symmetric Group (S_3) controllers providing invertible, loss-free data rollbacks.
7. **Graph Neural Network (GNN):** PyTorch Geometric modules processing large-scale empirical church network graphs.

2 Context Layer: Interlocking Decision Forests

2.1 Decision Trees as Associative Agent Memory

Decision trees function as the agent's structural memory matrix and are used directly for context classification. The forest learns complex relational concepts as presented by sparse multi-modal exemplars. The non-terminal internal nodes are derived dynamically from the exemplars, evaluating testing features or structured parthood (`part_of`) relationships.

The decision tree recursion is explicitly configured to support nested conceptual compositionality, enabling the system to learn high-level meta-concepts strictly in terms of simpler, underlying local sub-concepts. Concepts are formulated in LISP S-expressions and can be indefinitely nested. Terminal leaf nodes retain partial exemplar arrays and domain-specific solution techniques. There are multiple independent trees trained for each category of concepts, ensuring that structurally analogous scenarios converge cleanly onto the exact same leaf node.

2.2 `kb.pl` Logic as the Attention Guardrail

The Prolog logic layer operates as an explainable attention filter. If a decision tree attempts to shift its focus or generate a new, random feature split to minimize statistical error over noisy data, the `kb.pl` file audits that choice. It blocks the tree from chasing noise or hallucinating branches, verifying that variables are legally and structurally meaningful to the macro-institution.

3 Training and Inferencing Plane

3.1 Reduced-Vocabulary Intrusion Detection Gate

To protect the system from un-vetted adversarial payloads, the LLM provides all prompts in a strictly reduced vocabulary form to the neuro-symbolic intrusion detection system (IDS). This system was engineered in a previous development phase; it is maintained as a decoupled, upstream filter boundary to allow current research to focus exclusively on multi-agent system (MAS) consensus networks and parallel branching poset architectures without character cross-talk.

3.2 Inductive Ontology Replacement and Naming Realignment

Traditional data systems struggle with rigid ontology alignment when merging separate agent theories. Our architecture replaces manual ontology revision entirely by utilizing the unique mathematical properties of learned decision trees, rendering the shared strategy completely *inductively definable*.

The 5-step merging procedure unrolls at the target agent as follows:

1. **Logic Basis Consensus:** The target agent checks and achieves logical consensus on the received `kb.pl` file across the branching poset tier.

2. **Model Payload Buffering:** The target node copies the incoming abstract decision tree from the source agent directly into local memory.

3. **Logical Interrogation and Repair:** The target agent unrolls its local complex exemplars stored within its pre-resident `exs.pl` file. It strips their extensional type classification names, converting them into blind, intensional queries sent to both its local abstract tree and the copied source tree. If a nomenclature variance occurs between the tree outputs, the target logs

the discrepancy, captures the source agent's naming convention, and appends the corrected pattern to its local exemplar list.

4. **Inductive Retraining Pass:** Once the single-pass comparison completes, the target agent executes an inductive learning run over the newly updated, repaired `exs.pl` file matrix.

5. **Ontology Substitution:** The newly trained tree cleanly replaces the older model, serving as a pristine, conflict-free semantic merger.

3.3 Cross-Domain Isomorphism and Typological Analogies

We scale this architecture to support cross-domain reasoning by analogy, demonstrating that the structural form of an optimization problem applies with identical integrity across physical, social, and spiritual lattices. When the intensional type-label stripping operator purges localized name tags, disparate realities map to identical geometric paths within the decision forest:

- **Physical Grid Domain:** Detects synchronized AI data center load episodes and cross-substation power surge correlations to prevent transformer thermal melt perimeters.
- **Socio-Pragmatic Domain:** Parses recursive level-k strategic deception loops or John Searle's institutional status assignments (X counts as Y in C).
- **Scriptural-Spiritual Domain:** Evaluates salvific extraction trajectories by tracing historical typologies, mapping the relational structure of Judith's mountain crevice rescue directly to the non-convex trap coordinates of the Prodigal Son.

By unifying these domains, the system reveals that a physical power-grid crisis is structurally isomorphic to a strategic deception or a spiritual extraction. God's perspective operates as the ultimate atemporal sovereign controller (W_2) at the apex root node ($\top$), breaking the closed syntax limits of local human observation planes to enforce absolute safe performance perimeters.

4 Conclusion and Future Horizons

We have presented a robust, unified Neuro-Symbolic AI architecture that successfully scales across continuous physical motion, recursive theory of mind lattices, and discrete Searle institutional speech acts. By grounding continuous neural transformations into explainable symbolic guardrails, the framework eliminates VRAM memory black-box opacity and prevents hallucinatory logic paths. Future work will expand the algebraic solver constraints to track massive distributed multi-graph consensus splits in real time over volatile network nodes.

5 References

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